

The effect of light-induced plasma on propagation of intense fs laser radiation in c-Si

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Abstract The strong two-photon absorption and induced plasma are well-known factors restricting the penetration of intense focused laser pulses in silicon crystals. In this paper, we demonstrate that the role of plasma is not exhausted by the defocusing effect in a beam waist. We investigated experimentally the propagation of an IR laser pulse focused inside a silicon target and found a complex filament-like transformation of the beam. The results of numerical simulation suggest that this behavior is the result of wavefront distortion induced by the subsurface plasma ρ_s .

1 Introduction

Laser irradiation is a conventional technique for modifying the material structure inside the bulk of transparent materials, both amorphous [1, 2] and crystalline [3, 4]. For any lattice rearrangement inside a solid to succeed, sufficient energy should be transferred inside and deposited there. The use of lasers for this purpose is generally accepted, as intense light via nonlinear absorption readily causes ionization, which, in turn, can be followed by a change in the lattice structure. Usually the concentration of carriers induced in solids by the oscillating field of a modern light source can be very high and easily exceed the magnitude of $N_c \sim 10^{21} \text{ cm}^{-3}$, when optical breakdown occurs and the lattice is destroyed.

It is to be stressed that in the case of wide band-gap solids, the optical energy can be deposited in a well-localized region near a beam waist, thus ensuring high spatial resolution of laser machining. The conventional way to spatially concentrate light is focusing. An optical lens transforms the original nearly plane wavefront of a laser beam into a spherical one, making it converge after passing through the lens. Diffraction is the only phenomenon, which delimits the light propagation in the linear case and prevents the collapse of the focused beam. Light is localized in the Rayleigh region with a radius of $w \sim \lambda/NA_{\text{beam}}$, where λ is the wavelength and NA_{beam} characterizes the focusing angle.

The necessity of locally ionizing a substance prompts us to use short-pulse irradiation with peak intensity $> 10^{12} \text{ W cm}^{-2}$, which triggers a number of nonlinear processes. In terms of the problem of dense plasma creation in the bulk of a solid, the nonlinear light–matter interaction can be described by considering mainly two physical effects: the optical Kerr effect and plasma defocusing. The first one acts cumulatively against the diffraction divergence and tries to collapse the beam. The reason the second effect is decreasing in the local refractive index n is the emergence of free carriers. So, laser-induced ionization is a self-restricted process: The denser is the plasma, the less pellucid it is. It is to be noted, however, that the optical nonlinearity in most of the centrosymmetric media (like fused silica and diamonds) is relatively low in the NIR range, and the described effects come into play if the light intensity is high enough. In other words, practically the nonlinearity has to be considered as only being inside the beam waist.

Crystalline silicon (c-Si), which is of great interest in terms of its bulk treatment, has a quite different response to

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intense light irradiation. Evidently, the possibility of bulk nonlinear ionization necessitates that the laser wavelength lies in the region below the band gap, where the crystal is transparent. Most of the known attempts to modify the structure of c-Si by intense laser irradiation were performed with $\lambda = 1.2\ \mu\text{m}$ that is, under two-photon absorption (TPA) conditions [5–10]. These experiments demonstrated the strong effect of the optical intensity limiting that, in turn, restricts the ionization in the bulk of silicon.

Two main processes lead to what is referred to in [5] as light delocalization. Firstly, the strong TPA makes the dissipated energy spread over all the beam path, including the region beyond the Rayleigh range (the pre-focal region), where the light propagation follows geometrical optics. As a result, TPA substantially depletes the laser fluence F while a laser pulse with original energy E reaches the Rayleigh region, with the dependence $F(E)$ inside the laser waist being approximately logarithmic [11].

Another process that prevents light from concentrating in the focal region is the formation of e–h plasma, which is at the origin of the beam defocusing. Despite the fact that the e–h plasma observed in irradiated silicon is much less dense than in wide gap insulators [12], the defocusing effect in c-Si is well-pronounced because of the small effective mass of the free carriers.

It is necessary to note that the influence of the light-induced plasma on the propagation of an intense laser pulse is usually explored near the focal region. The pre-focal region still remains uninvestigated. Meanwhile, huge intensity limiting in c-Si suggests that the ionization there is strong enough. Then, the pre-focal plasma could be, at least in principle, dense enough to markedly contribute to the beam distortion. In order to investigate this possibility, we explore it both experimentally and theoretically. The experimental study revealed that the laser beam distortion in c-Si appears to be more complex than the traditional picture of femtosecond beam propagation in solids. The numerical simulation performed allowed establishing that the pre-focal plasma is indeed a key factor affecting the optical field both inside and beyond the Rayleigh region.

2 Experimental results

The employed laser system was based on a Ti:Al₂O₃ oscillator (Tsunami) and a regenerative amplifier (Spitfire, Spectra Physics). A parametric amplifier (OPA-800, Spectra Physics) was used to convert 800 nm radiation into $\lambda = 1.2\ \mu\text{m}$. IR pulses with duration $\tau_{\text{FWHM}} \approx 250$ fs and maximum energy $E = 100\ \mu\text{J}$ were focused by an aspherical lens ($f = 16$ mm, $NA_{\text{beam}} = 0.06$) inside the c-Si sample ($30 \times 9 \times 3$ mm) along the Z-axis (Fig. 1a).

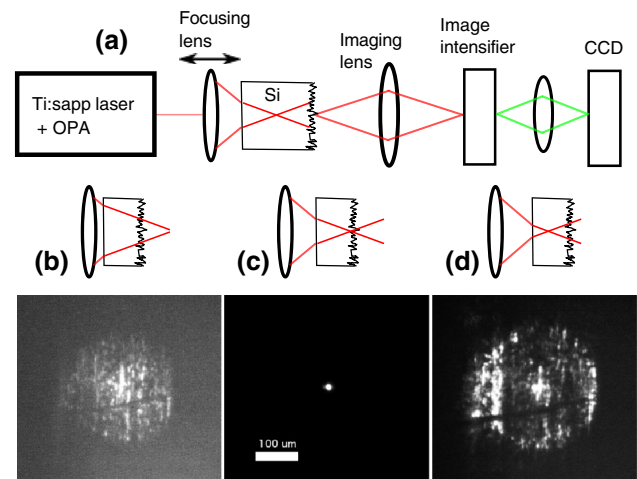


Fig. 1 **a** The scheme of optical measurements; **b–d** the focus positions and corresponding distributions of laser fluence on the sample backside ($E = 50\ \mu\text{J}$)

In order to characterize the laser fluence distribution inside the sample, a simple method resembling the Z-scan technique was applied. The focusing lens was moved along the Z-axis in such a way as to put the geometrical focus position sequentially inside and after the 9-mm-thick sample. The laser beam passes through the crystal and illuminates the backside, forming certain spatial patterns, which are strongly dependent on the position of the focusing lens and on the energy of the pulse. Care was taken so that the distribution of the local irradiance on this surface was measured accurately. For this purpose, the preliminarily mechanically scratched backside of the sample was projected on the CCD matrix via an image intensifier tube. Diffusively scattered light ensured the exact detection of the irradiance pattern. The photographs taken contained information about the distribution of the laser fluence F in a plane perpendicular to the Z-direction. A calibration procedure performed for the linear propagation case (for very small E) allows us to calculate the absolute value of the local F .

Figure 1b–d shows the photographs demonstrating the variations of the laser fluence along the beam while the pulse ($E = 50\ \mu\text{J}$) propagates through the silicon bulk. If the focus lies behind the sample, the nonlinear distortions are relatively small and the observed F distribution is close to Gaussian (Fig. 1b). When the focus comes to a point near the backside, the light converges to a small spot, also not revealing any remarkable nonlinear beam transformation (Fig. 1c). This occurs only when the focus occurs inside the crystal (Fig. 1d). In order to clarify the small-scale structure of the F distribution, we collected a number of CCD images by scanning the sample in the $X - Y$ plane. The data obtained (Fig. 2) show that the bell-shaped fluence profile falls apart as soon as light passes the focal point.

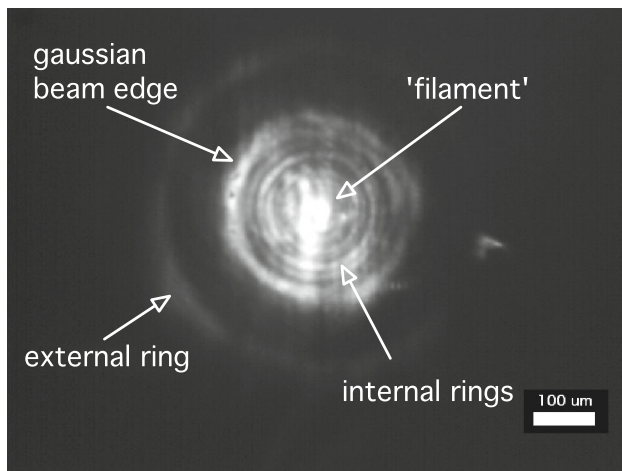


Fig. 2 Fine structure of distributions of laser fluence on the sample backside

The beam separates into distinct individual parts. Firstly, a prolonged filament appears in the center of the beam. Secondly, there is observed a clear-cut stable ring, which increases in diameter as the distance between the focus and the backside plane increases. This is nothing but the wings of the original Gaussian beam, which are relatively low in intensity and, thus, likely to not undergo any nonlinear rupture. Also, a family of a few (from one to five) much less bright internal and external rings can be detected in the pattern. These rings vary in size in a complex way, depending on the pulse energy. Moreover, the role of the pulse energy is crucial: The external rings were observed if and only if the pulse energy exceeds a certain threshold $E > E_s \approx 10 \mu\text{J}$. On the other hand, the beam kept its bell shape while $E < E_f \approx 1 \mu\text{J}$.

Figure 3 summarizes the observed behavior of pulses launched in Si bulk, in dependence on their energy. The geometrical parameters of the beam are illustrated in the upper panel (Fig. 3a), which depicts the radius of the outer constituent w_o (Gaussian wings) and the radius of the filament-like constituent w_f for $E = 50 \mu\text{J}$. The dependence of $w_o(z_{\text{opt}})$ appears to be well approximated by the well-known model of linear propagation of a convergent Gaussian beam with a waist radius of $w_o(0) = 6.5 \mu\text{m}$. On the other hand, the diameter w_f was found to be roughly constant $\approx 10 \mu\text{m}$ for a distance of slightly over 0.5 cm.

Figure 3b shows the alteration of the local laser fluence along the Z-axis for different $E = 0.01 - 50 \mu\text{J}$. It can be seen that the curves $F(z_{\text{opt}})$ are considerably non-symmetrical, and this asymmetry tends to increase with an increase in E . Furthermore, the maximum value F_{max} clearly saturates as E grows. Both the effects result obviously from beam attenuation due to the huge TPA-induced

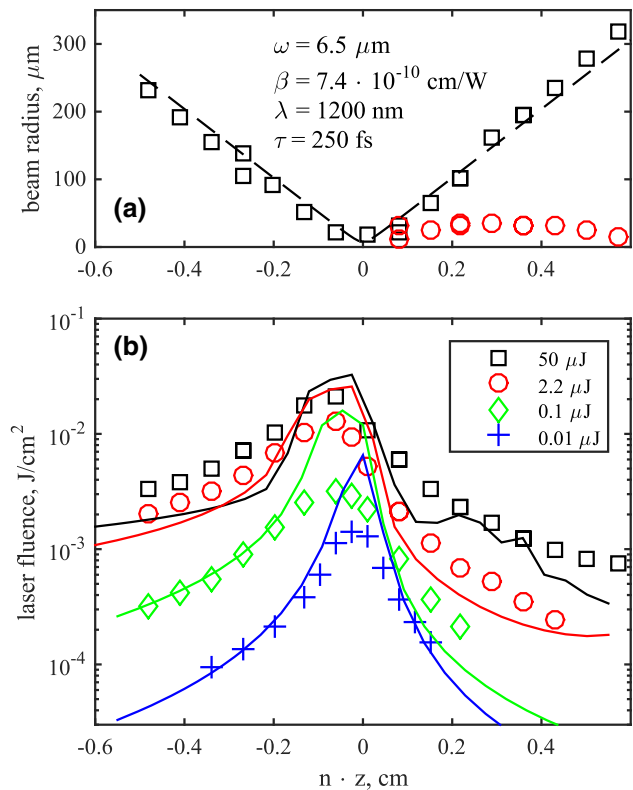


Fig. 3 **a** The radii of different constituents of a laser beam and **b** the laser fluence on the optical axis versus $z_{\text{opt}} = zn$, where z is the relative position of the focusing lens and $n = 3.5$ is the refractive index of c-Si. The symbols are experimental data, while the lines are taken from a numerical simulation (see the next section)

losses. However, the peak power of the radiation falls below the critical level ($P_c \approx 13 \text{ kW}$ for silicon) only for the smallest energy $E = 0.01 \mu\text{J}$. Indeed, inside the Rayleigh range, the value of the laser fluence corresponding to P_c is $F_c \approx P_c \tau / \pi w^2 \approx 2 \cdot 10^{-3} \text{ J}/\text{cm}^2$. One can see that $F_{\text{max}} > F_c$ even for $E = 0.1 \mu\text{J}$ (Fig. 3b).

This observation suggests ruling out the conventional mechanism of filament formation. Usually, the filamentation is attributed to a cycling process of focusing–defocusing, which, in turn, is the result of subtle competitions between Kerr and plasma-induced nonlinearities. The filamentation regime can be reached when the peak power is larger than P_c and self-focusing occurs upon pulse propagation. However, in our experiments, the light power in the focal region is quite enough to form such a filament not only when it appears ($E > E_f$), but also when it does not ($E = 0.1 \mu\text{J} < E_f$).

What then is responsible for the complex light pattern revealed in the region behind the focal point? To investigate the spatial beam splitting, a computer simulation of femtosecond pulse propagation in c-Si target was performed.

3 Numerical simulation

The scalar equation of nonlinear Schrodinger type describing the evolution of an applied optical field $\mathcal{E}$ is

$$\frac{\partial \mathcal{E}}{\partial z} = \frac{i}{2k} \left(\frac{1}{r} \frac{\partial}{\partial r} + \frac{\partial^2}{\partial r^2} \right) \mathcal{E} + \frac{ikn_2}{n_0} |\mathcal{E}|^2 \mathcal{E} - \frac{i2\pi k e^2}{n_0 \omega^2 m} \rho_e \mathcal{E} - \frac{1}{2} \beta |\mathcal{E}|^2 \mathcal{E} \quad (1)$$

The first term in the right part of (1) is responsible for the diffraction of linearly polarized light. The approximation of a slowly varying envelope allows taking into account the induced dynamics of optical polarization caused by both the third-order nonlinearity of optical susceptibility—the Kerr effect—the second term in Eq. (1) and e–h plasma formation (the third term). The fourth term takes into account light losses caused by TPA. The contribution of inverse bremsstrahlung absorption was omitted because previous experiments showed TPA to be the dominant mechanism responsible for optical ionization in silicon [13]. The relaxation of the plasma was also supposed to be negligible, as the relaxation time in c:Si (~ 10 ns) is much higher than the pulse duration. The expression for the concentration ρ of the e–h plasma is therefore very simple:

$$\frac{\partial \rho}{\partial t} = \frac{\beta |\mathcal{E}|^4}{2\hbar\omega} \quad (2)$$

Equation (1) was numerically solved using a finite difference method. The selection of the region of interest inside the silicon target was defined by the experimental conditions described above. Due to the strong TPA, the plasma cloud was expected to be smeared out over all the pathway of light propagation. This is the reason why the solution was traced along a distance of 9 mm, whereas most of the previous simulations only treat an optical field near the focus.

In order to ensure a reasonable rate for the process of calculation and get a solution in the range from -4.5 to 4.5 mm along the Z -axis, two assumptions were tried. Both the term responsible for the group velocity dispersion and the operators responsible for the self-steepening and the space–time coupling [14] were estimated, and their influence on the solution was concluded to be almost negligible.

The initial parameters of the radiation were taken from the experiment. The pulse profile was approximated by a Gaussian both in space and in time, with a waist radius at the $1/e^2$ intensity level of $w = 6.5 \mu\text{m}$ and a duration of $\tau_{\text{FWHM}} = 250$ fs. The following material constants were used: The refractive index of c-Si is $n = 3.5$, the Kerr coefficient is $n_2 = 2.6 \cdot 10^{-14}$, the TPA coefficient is $\beta = 7.4 \cdot 10^{-10} \text{ cm/W}$ [15], and the reduced optical mass

m of an excited carrier in Si was supposed to be 0.3 times the free electron mass [16].

The calculated axial fluence is depicted in Fig. 3. One can see that it approaches the measured data, and what is even more important, properly describes the measured energy depletion along the Z -axis due to TPA. A pronounced gap between the calculated and observed F appears in the focal region. This results, apparently, from an experimental underestimate of the size of the beam, which, in comparison with the applied Gaussian profile, dramatically diminishes due to self-focusing.

The self-focusing effect is clearly seen in Fig. 4, which shows the calculated laser fluence inside the silicon target for all energies concerned. In order to illustrate all the features of the light propagation in silicon, we plotted $\log(F)$ presented in arbitrary units. The data in Fig. 4 are normalized to a maximum $F_{\text{max}} = 0.38 \text{ J/cm}^2$, which was found definitely for the $50 \mu\text{J}$ pulse.

As seen in the plot, the laser beam becomes distorted even at $E = 0.1 \mu\text{J}$ (Fig. 4). The surface plasma is very small yet $\rho_s \approx 3 \cdot 10^{15} \text{ cm}^{-3}$. As the optical energy is mostly localized in the Rayleigh range, the beam starts interacting with the bulk plasma $\rho_b \approx 5 \cdot 10^{18} \text{ cm}^{-3}$, diffracting on it and splitting behind the focus: There are intensity distribution changes in the far-field region. The pattern does not much change for a

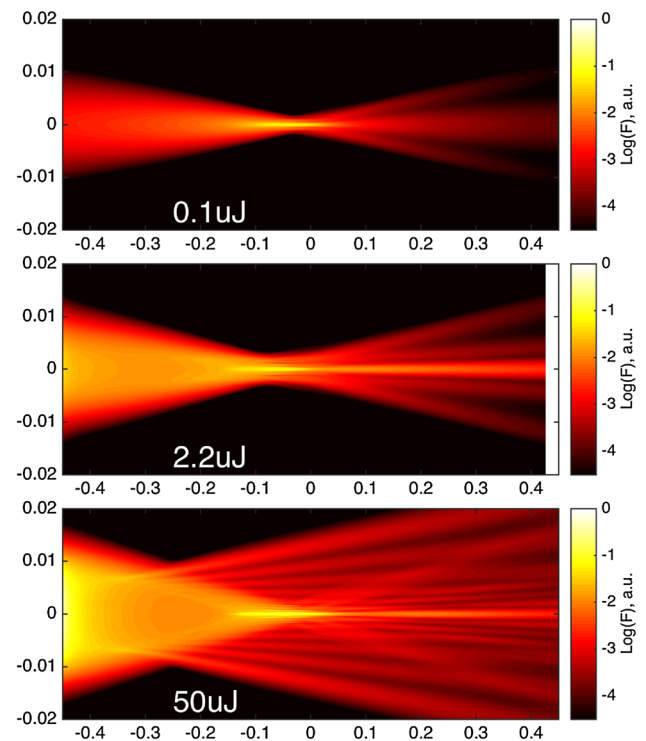


Fig. 4 2D maps of laser fluence distribution inside the c-Si target for varied pulse energies. Light moves *left to right*. z and r values are given in microns

pulse of $E = 2.2 \mu\text{J}$. The bulk plasma growth saturates at the level $\rho_b \approx 5 \cdot 10^{18} \text{ cm}^{-3}$. Even though surface plasma ρ_s drastically rises to $\approx 2 \cdot 10^{18} \text{ cm}^{-3}$, it evidently does not much affect the light propagation.

Nevertheless, the beam was found to be completely transformed for a pulse of $50 \mu\text{J}$, when $\rho_s \approx 10^{21} \text{ cm}^{-3}$ appears to be much denser than $\rho_b \approx 10^{19} \text{ cm}^{-3}$. The solution points to the fact that the beam splitting is initiated exactly in the sub-surface region. A ring-like structure is developed, as shown in Fig. 4, long before the focusing point, contradicting the experiment. Indeed, a similar intensity distribution was found in the far-field region, but not in the focal point.

This apparently contradictory observation can be reconciled by noting that at $z = 0 \mu\text{m}$ the ratio between the fluencies in the central and the marginal rays is about > 100 . The employed detectors could not take such a contrast picture correctly. But then TPA depletes the power in the central ray and at $z = +200 \mu\text{m}$ the intensity ratio decreases to ~ 10 , clarifying a ring-like structure of the laser beam. In fact, the calculated pattern correlates perfectly with the measured one, demonstrating, in particular, the external ring that is larger than the 'wings' of the Gaussian beam (see Fig. 2).

The presented results suggest that the subsurface plasma plays a key role in the laser beam splitting. This assumption agrees also with the evolution of the light-induced processes in silicon upon laser pulse propagation (Fig. 5). The point is that the dynamics of the optical energy dissipation are quite different in the bulk from what they are near the surface.

The forward part of the pulse generates free carriers at the focal point much faster. The immediate cause is that in the beginning of the exposure, the local intensity in the waist is much higher than on the surface. The denser the plasma becomes, the stronger the defocusing is, making the light propagate the other way and limiting the local concentration of the carriers.

The calculations (Fig. 5) indicate a clear-cut delay between the moments when ρ_s and ρ_b reach a saturation level. The dynamics of the beam transformation is exactly the same: The beam splitting induced by the surface plasma is delayed $\sim 150 \text{ fs}$. It appears after $\sim t = -50 \text{ fs}$, while the distinct effect of the beam diffraction on the bulk plasma can be found at $\sim t = -200 \text{ fs}$. Note also that after $t \approx 0 \text{ fs}$, the intensity pattern stabilizes and does not change drastically (only the brightness falls) throughout the rest of the pulse.

4 Discussion

So, the modeling of the propagation of femtosecond light through a silicon crystal proves the formation of subsurface plasma to be followed by the splitting beam. The calculated

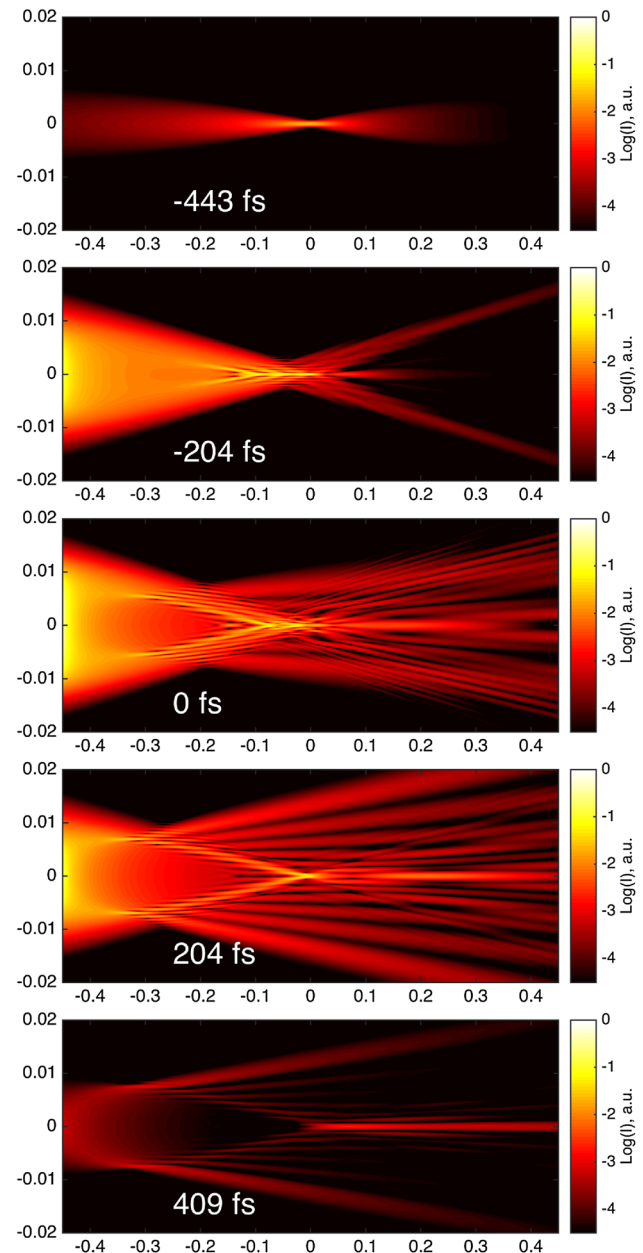


Fig. 5 2D maps of laser intensity distribution inside the c-Si target for $E = 50 \mu\text{J}$

pattern this splitting produces is rather complex and, to account for the overall features of the light field distribution, an intuitive model of the effect of the surface plasma on the beam has been given. Besides accounting for the behavior of the beam, the model allows deducing a simple formula for the critical threshold of laser fluence at which this behavior starts.

First of all, it is necessary to emphasize that in contrast to focal plasma, the concentration of subsurface e-h pairs produced in the first laser pulse is limited only by the plasma mirror phenomenon, and thus, it can reach the huge value of $\rho_e \approx 2 \cdot 10^{22} \text{ cm}^{-3}$. Then, the subsurface

plasma spreads relatively deeply into the medium: for $E = 50 \mu\text{J}$, the depth at which ρ falls to 10 % of its value just beneath the surface was evaluated to be $\approx 20 \mu\text{m}$. It is of interest that such a deeply penetrating plasma is likely to modify the crystal structure. The possibility of fs laser-induced inscription of waveguides in Si was demonstrated in [17]. They pointed out a strange fact: the lattice modification appeared at only small distances $< 20 \mu\text{m}$ below the surface, irrespective of the focusing depths. Such a behavior can be attributed to the subsurface plasma concerned here.

Returning to the previous point, the subsurface layer of dense plasma $\rho < \rho_c$ obviously substantially affects the trailing part of pulse. The plasma layer acts as a thin negative lens, and the resulting phase of the wavefront is defined as the sum of spherical and Gaussian components:

$$\phi(r) = \frac{D}{\lambda} \cdot \exp\left(-\frac{r^2}{w^2}\right) + \frac{kr^2}{2R} \quad (3)$$

where $k = 2\pi/\lambda$, D denotes the optical thickness of lens, w is the Gaussian radius of the beam on the sample surface, and R is the radius of curvature of the original beam wavefront.

The induced wavefront distortion is depicted schematically in Fig. 6a. The modeling of the propagation of a Gaussian beam in the framework of this model was done as well, and the laser fluence map $F(r, z)$ is plotted in Fig. 6b. This calculation is the solution of Eq. (1) for the wave (3), excluding all the nonlinear terms. Neither TPA losses, nor the Kerr effect, nor optical ionization was taken into account. All the plasma was simulated by means of the surface negative lens with $D = 4\lambda$.

It is easy to see that despite such a simplification, the pattern in Fig. 6b is quite close to the pattern in Fig. 4 ($50 \mu\text{J}$), and thus, the subsurface plasma is proven to be the key factor driving the laser beam splitting in silicon.

The distorted wavefront changes, in particular, the superposition of waves coming from the crystal surface and causes an evident self-interference inside the caustic. As a schematic ray tracing shows (Fig. 6a), the plasma lens produces new marginal rays. Besides the converging rays (1) originating from external focusing, the divergent rays (2) emerge. In the first approximation, they start from the point $r = \frac{w}{\sqrt{2}}$ where $\frac{\partial \phi}{\partial r}$ reaches its maximum. The marginal rays divide the caustic into three parts: A, B, and C (Fig. 6a). Simple ray tracing shows that each of the regions features an inherent interference pattern. Region A does not contain any superposition of different waves, which results in the absence of interference inside. In contrast, two/three rays are capable of crossing inside region B/C, creating specific interferometric patterns with remarkably different periodicities.

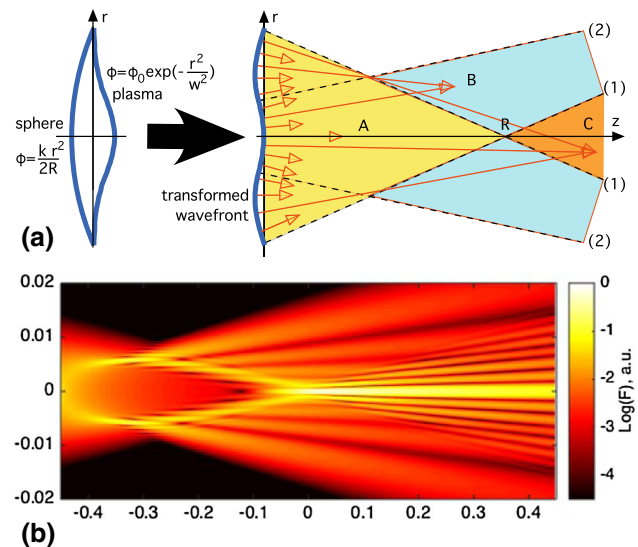


Fig. 6 **a** Surface plasma-induced distortion of a focused laser beam and the interference pattern that appeared. The *dotted lines* are marginal rays; **b** computer modeling of propagation of focused beam through a negative lens with a Gaussian profile and optical thickness $D = 4\lambda$ at the central point $r = 0 \mu\text{m}$

The waist of the Gaussian beam is also hard reshaped. The huge longitudinal aberration lengthens the main focus region, resulting in a seemingly filament-like behavior of the beam (Fig. 6b). Certainly, in the nonlinear case, the TPA continuously depletes the light in the regions of optical field localization.

As was mentioned above, the model allows us to estimate the threshold value F_s below which no beam splitting is observed. The criterion of the onset of defocusing can be written as

$$\frac{\partial^2 \phi}{\partial r^2} \Big|_{r=0} = 0 \quad (4)$$

where ϕ is defined by (3). From the physical point of view, this condition means the flatness of the wavefront on the beam axis ($r = 0$). Equations (4) and (3) give

$$D_s = \lambda w^2 k / 8R \approx 1.5\lambda \quad (5)$$

For beam splitting to occur, it is therefore necessary that the optical thickness of the subsurface plasma layer exceeds $(1 \div 2)\lambda$. On the other hand, the intensity I of light propagating in the TPA medium is attenuated with an increase in distance Δz as: $I(z) = I_s / (1 + I_s \beta \Delta z)$, where I_s denotes the intensity in the plane of the surface. Consequently, the local concentration of free e-h carriers is $\rho(z) = I^2 \beta \tau / 2\hbar\omega$. Taking into account the Drude model, the optical depth of plasma layer can be expressed as

$$D_s = \int \Delta n \, dz = \int \frac{e^2 \rho(z)}{2m\omega^2} \, dz = \frac{e^2}{2m\omega^2} \frac{F_s}{2\hbar\omega} \quad (6)$$

Taken together, (5) and (6) give an estimate for the threshold of surface laser fluence in our focusing conditions $F_s \approx 0.16 \text{ J/cm}^2$ and corresponding pulse energy $E_s \approx 30 \mu\text{J}$, which are in good accordance with the experimental and simulated data.

5 Conclusions

In summary, a new aspect of the interaction of intense laser pulses with a TPA medium, namely the formation of a subsurface plasma lens associated with the nonlinear ionization, has been investigated experimentally and theoretically. The appropriate model of the effect of the surface plasma on a focused femtosecond radiation has been proposed.

As well as the focal plasma, the surface plasma drives the light in tailoring part of the pulse. But, in contrast to focal plasma, which rather defocuses the light, the surface plasma is limited only by the plasma mirror effect and thus can reach the magnitude of $> 10^{22} \text{ cm}^{-3}$. As a result, the surface plasma distorts the focused beam to such an extent, that it is worth saying about beam splitting. The plasma-driven beam transformation is a rather harmful effect in terms of the possibility of laser treatment in the bulk of a TPA solid. Like the TPA-induced laser power depletion and focal plasma defocusing, it hinders the localization of the optical field inside a TPA medium.

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